

The Effects of Cognitive Behavior Therapy on Chronic Fatigue Syndrome and its Potential Moderators: A meta-analysis

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Objective: This meta-analysis attempt to assess quantitatively the effectiveness of cognitive behavioral therapy (CBT) based interventions for chronic fatigue symptom (CFS) adult patients. The research questions are listed as follow: (1) Does the reported CBT treatment effects differ across the three mostly used CFS severity measurements (CIS-F, CFQs, SF-36)? (2) Do the treatment effects of studies with within-group design significantly differ from that of between-group design? (3) Do gender, age, illness duration, publication year and treatment duration have any moderation effect to the CBT effect on CFS patients? (4) Does the data collection year moderate the CBT effect?

Methods: Nine databases were searched. Papers from 2002-2012, using the keywords “chronic fatigue syndrome”, “treatment” and “outcome”. Only those studies with (1) CBT treatment on CFS patients as well as (2) containing sufficient information for effect size calculation were taken into this study. After search, a total of 22 studies (sample size 3176, mean age of subjects is 40.14 years old) were included in this meta-analysis. Lipsey and Wilson’s (2001) macros for SPSS were used in the data analysis.

Results: The main results can be summarized as follow: (1) The analyses showed that CBT intervention possess a promising effect in terms of CFS treatment. Also, the reported treatment effects of CBT intervention in CFS patients varied significantly among the three kinds of measurement. (2) The studies with between-group and within-group design showed no significant but yet marginal effect size difference. The average effect size of intra-group design studies is higher than that of inter-group design studies. (3) Patient’s gender, age, illness duration, treatment duration, year of data collection showed no moderation effect to this kind of treatment process. (4) However, the treatments of 2008 to 2012 showed a significantly better treatment effect than that of 2002 to 2007. This finding suggests that the advance of CBT technique as well as the latest understanding of CFS prognosis may contribute to the better off treatment effect.

Keywords: *chronic fatigue symptom, cognitive behavioral therapy, meta-analysis*

* All four authors have equal contribution to this study. Their names are sorted by alphabetical order.